Electrochemical Device Modelling

End-to-End Computational Model of the Multiplexed Wound Protease Cartridge

Eton iGEM 2026, Dry Lab

MMP-9 / HNE Ratiometric SWV Sensor

Scope and motivation. There is no physical prototype of the electrochemical cartridge. Every physical process from wound-fluid biology through to the clinical readout number must be modelled mathematically and computationally. The chain is fully coupled: outputs of each layer feed as inputs to the next. The model must produce falsifiable, quantitative predictions (predicted signal F_k vs known protease concentration) that can be validated against wet-lab calibration data, satisfying the iGEM gold criterion that models predict specific experimental outcomes that wet lab then confirms.
This document enumerates every sub-component that requires explicit mathematical treatment across 16 modelling layers, with the governing equations, key parameters, and inter-layer coupling for each.

Layer 0: Wound biology (system input)

The model input is the active concentration of MMP-9 and HNE in wound fluid at the time of sampling. These are not directly observable; they must be derived from the wound biochemical ODE system (Pillar 2, `report_project_proposal.tex`).

ODE system for protease activation

$$\frac{d[\text{proMMP-9}]}{dt} = -k_{\text{act}}[\text{plasmin}][\text{proMMP-9}] + k_{\text{syn}} \tag{1}$$

$$\frac{d[\text{MMP-9}]}{dt} = k_{\text{act}}[\text{plasmin}][\text{proMMP-9}] - k_{\text{TIMP}}[\text{TIMP-1}][\text{MMP-9}] \tag{2}$$

$$\frac{d[\text{HNE}]}{dt} = k_{\text{rel}} - k_{\text{A1AT}}[\text{A1AT}][\text{HNE}] \tag{3}$$

Outputs: $[\text{MMP-9}]_{\text{active}}$ and $[\text{HNE}]_{\text{active}}$ as functions of wound state and time since debridement.

Environmental parameters

- **pH**: range 6.5–8.2 (chronic to acute). Affects enzyme k_{cat} via pH-activity profile, formal potential E°_{MB} (shifts $-59\,\text{mV/pH}$ for the $2e^-/2H^+$ MB couple), and Debye screening.
- **Ionic strength** I: physiological $\approx 150\,\text{mM}$. Debye length $\lambda_D = \sqrt{\varepsilon\varepsilon_0 RT/(2F^2 I)} \approx 0.8\,\text{nm}$.
- **Temperature**: operating range 20–37°C. Every rate constant downstream is temperature-dependent.
- **Oxidative species**: H_2O_2, hypochlorite can oxidise the Cys–Au thiol bond. Model as first-order SAM degradation: $d\Gamma/dt|_{\text{ox}} = -k_{\text{ox}}[\text{ROS}]\Gamma$.
- **Matrix proteins**: albumin, fibrinogen contribute to non-specific adsorption (NSA) on gold. NSA fraction θ_{NSA} blocks probe-electrode area and increases baseline capacitance.
- **Interfering proteases**: cathepsins B/D/G, plasmin, urokinase. Each contributes off-target cleavage of the probe substrates with its own $k_{\text{cat,j}}$ and $K_{m,j}$ (from MEROPS). Must be included as additive cleavage terms.

Layer 1: Sample collection and pre-analytical

Swab absorption and elution

Let V_{abs} = volume absorbed by swab (20–200 μL), η_{elut} = elution efficiency (0.4–0.8 for proteins, size-dependent).

Concentration in eluate: $[E]_{\text{eluate}} = [E]_{\text{wound}} \times \eta_{\text{elut}}$.

Analyte stability during transit

First-order degradation from collection to assay (time t_{transit}):

$$[E]_{\text{eluate}}(t_{\text{transit}}) = [E]_{\text{eluate}}(0) \times \exp(-k_{\text{deg}}\, t_{\text{transit}}) \tag{4}$$

k_{deg} from literature self-digestion rates or measured stability assay.

Net dilution into cartridge

$$[E]_{\text{chamber}} = [E]_{\text{wound}} \times \eta_{\text{elut}} \times \exp(-k_{\text{deg}} t_{\text{transit}}) \times D_f \tag{5}$$

where $D_f = V_{\text{eluate}}/V_{\text{chamber}}$ is the dilution factor.

Inhibitor co-elution

TIMP-1 (28 kDa), A1AT (52 kDa), and SLPI (11 kDa) co-elute alongside the target enzymes with size-dependent efficiencies $\eta_{\text{elut,I}}$. Inside the device they continue to suppress free enzyme during the 30-min incubation. The inhibitor concentrations entering the chamber are:

$$[I]_{\text{chamber}} = [I]_{\text{wound}} \times \eta_{\text{elut,I}} \times D_f \tag{6}$$

SLPI (11 kDa) elutes more efficiently than HNE (30 kDa), so the [SLPI]/[HNE] ratio in the chamber is higher than in wound fluid. These concentrations are passed to Layer 3 for in-device inhibition kinetics.

Sample pH after elution buffer mixing

The sample pH in the chamber is not simply the elution buffer pH. If wound pH deviates significantly from buffer pH:

$$\text{pH}_{\text{chamber}} \approx -\log_{10}\left(\frac{V_{\text{wound}} \times 10^{-\text{pH}_{\text{wound}}} + V_{\text{buffer}} \times 10^{-\text{pH}_{\text{buffer}}}}{V_{\text{wound}} + V_{\text{buffer}}}\right) \tag{7}$$

For an infected wound (pH 6.5) and PBS elution buffer: $\text{pH}_{\text{chamber}}$ can be as low as 6.9–7.1. This shifts E°_{MB} by up to 30 mV and reduces enzyme k_{cat} via its pH-activity profile. $\text{pH}_{\text{chamber}}$ (not 7.4) must propagate to Layers 6, 10, and 12.

Co-eluted redox-active species: glutathione ($[\text{GSH}]_{\text{wound}} \sim 0.1$–1 mM), ascorbate ($\sim 0.1$ mM), H_2O_2 (10–100 μM). These are tracked as $[\text{GSH}]_{\text{chamber}}$ for downstream MB oxidation state modelling (Layer 7).

Layer 2: Cartridge inlet and capillary fill

Chamber geometry

Chamber dimensions: 20×5 mm footprint, height $h \in [0.25, 0.50]$ mm. Volume $V = 100\,\text{mm}^2 \times h \approx 25$–$50\,\mu$L.

Capillary pressure and Lucas-Washburn fill

Capillary driving pressure across a rectangular channel ($h \ll w$):

$$\Delta P_{\text{cap}} = \gamma \cos\theta \left(\frac{1}{h} + \frac{1}{w}\right) \tag{8}$$

where γ is the sample surface tension (≈ 72 mN/m for aqueous, reduced by surfactants) and θ is the advancing contact angle (different for each surface: PDMS lid, gold floor, PETG walls).

Fill front position $L(t)$ by the Lucas-Washburn equation:

$$\frac{dL}{dt} = \frac{h^2 \Delta P_{\text{cap}}}{3\eta L} \quad \Rightarrow \quad L(t) = \sqrt{\frac{h^2 \Delta P_{\text{cap}}}{3\eta} \cdot 2t} \tag{9}$$

Fill fails if $\theta > 90^\circ$ on any surface (reverses ΔP_{cap}).

Air venting

Without a vent, trapped air pressure rises as L increases:

$$P_{\text{air}}(L) = P_{\text{atm}} \cdot \frac{L_{\text{chamber}}}{L_{\text{chamber}} - L} \tag{10}$$

Fill stalls when $\Delta P_{\text{cap}} = P_{\text{air}}(L) - P_{\text{atm}}$. Maximum fill fraction without vent:

$$f_{\max} = 1 - \frac{P_{\text{atm}}}{\Delta P_{\text{cap}} + P_{\text{atm}}} \tag{11}$$

A vent membrane with gas permeability P_{vent} must allow air escape at the same rate as sample advances.

Electrode exposure timing

Electrode at position x_i from inlet is wetted at time t_i when $L(t_i) = x_i$. For 30-min incubation: if $t_{\text{fill}}(x_8) - t_{\text{fill}}(x_1) \ll T_{\text{incubation}}$, all electrodes see effectively equal exposure. Quantify for chosen geometry.

Layer 3: Mass transport inside the chamber (incubation)

Post-fill transport regime

After filling, the device is static. Transport is purely diffusive. Péclet number during incubation: $\text{Pe} = 0$ (no flow).

Diffusion coefficients (Stokes-Einstein, r = hydrodynamic radius):

$$D_{\text{MMP-9}} \approx \frac{k_B T}{6\pi\eta r_{\text{MMP-9}}} \approx 5 \times 10^{-11}\,\text{m}^2\text{s}^{-1} \quad (M_W = 92\,\text{kDa},\ r \approx 3.5\,\text{nm}) \tag{12}$$

$$D_{\text{HNE}} \approx 9 \times 10^{-11}\,\text{m}^2\text{s}^{-1} \quad (M_W = 30\,\text{kDa},\ r \approx 2.0\,\text{nm}) \tag{13}$$

$$D_{\text{frag}} \approx 3 \times 10^{-10}\,\text{m}^2\text{s}^{-1} \quad (\text{released MB peptide fragment},\ M_W \approx 1\,\text{kDa}) \tag{14}$$

Vertical mixing time

$t_{\text{mix}} \approx h^2/(2D)$. For $h = 0.375\,\text{mm}$, MMP-9: $t_{\text{mix}} \approx 1400\,\text{s}$. The channel is *not* well-mixed vertically during a 30-min incubation. A 1D diffusion PDE in the z-direction (perpendicular to the electrode) is required.

Coupled PDE-ODE system

Let $z \in [0, h]$ with $z = 0$ at the electrode surface.

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \quad z \in (0, h),\ t > 0 \tag{15}$$

$$[E](z, 0) = [E]_{\text{chamber}} \quad (\text{initial condition}) \tag{16}$$

$$D_E \frac{\partial [E]}{\partial z}\bigg|_{z=0} = \frac{k_{\text{cat}}[E](0,t)\,\Gamma}{K_{m,\text{surf}} + \Gamma} \quad (\text{Robin BC, flux to surface}) \tag{17}$$

$$\frac{\partial [E]}{\partial z}\bigg|_{z=h} = 0 \quad (\text{no-flux at lid}) \tag{18}$$

Equation (17) couples the bulk PDE to the surface kinetics ODE (Layer 9).

Damköhler number

$$\text{Da} = \frac{k_{\text{cat}}\Gamma_0 / K_{m,\text{surf}}}{D_E [E]_0 / h^2} \tag{19}$$

$\text{Da} \ll 1$: reaction-limited (desired). $\text{Da} \gg 1$: transport-limited (signal reports diffusion rate, not k_{cat}/K_m). Must verify $\text{Da} < 0.1$ at expected Γ_0 and $[E]_0$.

MB fragment clearance

Released fragment ($D_{\text{frag}} \approx 3 \times 10^{-10}\,\text{m}^2\text{s}^{-1}$) must diffuse away from the electrode before SWV measurement to prevent a bulk diffusion peak contaminating the surface-confined peak. Clearance time: $t_{\text{clear}} \approx \delta^2/D_{\text{frag}}$ where δ is the Nernst diffusion layer thickness (10–100 μm for quiescent solution). For $\delta = 50\,\mu\text{m}$: $t_{\text{clear}} \approx 8\,\text{s}$.

Electrode cross-talk

Lateral diffusion of released MB fragment between adjacent electrodes (separation $d_{\text{elec}} \sim 2\,\text{mm}$): $t_{\text{cross}} \approx d_{\text{elec}}^2/(2D_{\text{frag}}) \approx 7000\,\text{s}$. Cross-talk negligible for $T_{\text{incubation}} < 30\,\text{min}$ provided $d_{\text{elec}} > 1\,\text{mm}$.

In-device inhibitor binding kinetics

After fill, co-eluted TIMP-1 and A1AT/SLPI react with free enzyme in the bulk solution. This depletes $[E_{\text{free}}]$ over time independently of surface cleavage. The diffusion PDE must include reaction terms:

$$\frac{\partial [E_{\text{free}}]}{\partial t} = D_E \frac{\partial^2 [E_{\text{free}}]}{\partial z^2} - k_{\text{on},I}[E_{\text{free}}][I_{\text{free}}] \tag{20}$$

$$\frac{\partial [I_{\text{free}}]}{\partial t} = D_I \frac{\partial^2 [I_{\text{free}}]}{\partial z^2} - k_{\text{on},I}[E_{\text{free}}][I_{\text{free}}] \tag{21}$$

For TIMP-1/MMP-9: $k_{\text{on}} \approx 2 \times 10^5\,\text{M}^{-1}\text{s}^{-1}$, $K_i \approx 0.01\,\text{nM}$ (essentially irreversible). At $[\text{TIMP-1}]_{\text{chamber}} = 10\,\text{nM}$: pseudo-first-order inhibition half-life $\approx 350\,\text{s}$ ($\sim 6\,\text{min}$) — significant over a 30-min incubation.

For A1AT/HNE: $k_{\text{on}} \approx 2 \times 10^7\,\text{M}^{-1}\text{s}^{-1}$; at $[\text{A1AT}]_{\text{chamber}} = 1\,\mu\text{M}$, half-life $\approx 35\,\text{ms}$ — essentially instantaneous. The device measures the free HNE that escapes inhibition, which is the clinically relevant quantity. The Layer 9 Robin BC must use $[E_{\text{free}}](z = 0, t)$, not total $[E]$.

Layer 4: Gold electrode surface

Electrochemical roughness factor

True electrode area $A_{\text{true}} = R_f \times A_{\text{geo}}$ where $R_f \approx 1.1$–1.5 for smooth evaporated gold. Measured by integrating the gold oxide stripping charge: $R_f = Q_{\text{stripping}}/(Q_{\text{ref}} \times A_{\text{geo}})$ where $Q_{\text{ref}} = 390\,\mu\text{C}\,\text{cm}^{-2}$.

All surface densities Γ are in units of $\text{mol}\,\text{cm}_{\text{true}}^{-2}$.

Thin-film and contact resistance

Gold trace resistance: $R_{\text{Au}} = \rho_{\text{Au}} L/(wt)$ where $\rho_{\text{Au}} = 2.4 \times 10^{-8}\,\Omega\,\text{m}$. For $L = 5\,\text{mm}$, $w = 0.5\,\text{mm}$, $t = 100\,\text{nm}$: $R_{\text{Au}} \approx 2.4\,\Omega$. Uncompensated resistance $R_u = R_{\text{Au}} + R_{\text{contact}} + R_s$.

Layer 5: Reference and counter electrodes

On-chip Ag/AgCl reference

$$E_{\text{ref}} = E^{\circ}_{\text{AgCl/Ag}} - \frac{RT}{F} \ln(a_{\text{Cl}^-}) \tag{22}$$

At $[\text{Cl}^-] = 110\,\text{mM}$ (wound fluid): $E_{\text{ref}} \approx +0.198\,\text{V}$ vs NHE $\approx -0.005\,\text{V}$ vs standard Ag/AgCl.

Reference drift arises from Cl^- depletion at the reference junction. Acceptable drift: $< 5\,\text{mV}$ over 30-min assay to avoid measurable peak position shift. Model as: $\delta E_{\text{ref}}(t) = (RT/F) \times \delta \ln(a_{\text{Cl}}(t))$.

Counter electrode

CE must supply current to all WEs. Max SWV peak current per WE $\approx 100\,\text{nA}$; for 8 WEs: $I_{\text{CE,max}} \approx 800\,\text{nA}$. Require $A_{\text{CE}} > 10 \times A_{\text{WE}}$ (rule of thumb for non-limiting CE). Verify CE potential does not exceed O_2 evolution limit ($E > +0.97\,\text{V}$ vs Ag/AgCl) during anodic SWV half-cycles.

Layer 6: Electrical double layer and solution resistance

Solution resistance

$$R_s = \frac{\rho_{\text{sol}}\, d_{\text{WE-RE}}}{A_{\text{cell}}}, \quad \sigma_{\text{wound}} \approx 15\,\text{mS}\,\text{cm}^{-1} \tag{23}$$

Uncompensated iR drop during peak SWV current (100 nA): $\Delta E_{iR} = I_{\text{peak}} R_s \approx 3\,\mu\text{V}$, negligible. Verify for maximum expected current.

Double-layer capacitance

For SAM-modified gold: $C_{\text{dl}} \approx 1\text{–}5\,\mu\text{F}\,\text{cm}^{-2}$ (SAM acts as thin dielectric: $C_{\text{dl}} = \varepsilon_0 \varepsilon_{\text{SAM}}/d_{\text{SAM}}$). Capacitive background in SWV:

$$I_{\text{cap}} = 2 C_{\text{dl}} A\, \Delta E_{\text{SW}} f_{\text{SWV}} \tag{24}$$

For $C_{\text{dl}} = 2\,\mu\text{F}\,\text{cm}^{-2}$, $A = 1\,\text{mm}^2$, $\Delta E_{\text{SW}} = 25\,\text{mV}$, $f = 60\,\text{Hz}$: $I_{\text{cap}} \approx 60\,\text{nA}$. Faradaic MB peak must be $> I_{\text{cap}}$ to be detectable; this sets the minimum useful Γ.

Debye screening and charged residues

$\lambda_D \approx 0.8\,\text{nm}$ at $I = 150\,\text{mM}$. Probe tether length ($\sim 5\,\text{nm}$) extends far beyond λ_D. For IPRTLTAA, the Arg residue (R, pKa 12.5) carries +1 charge at pH 7.4 and is exposed in the diffuse layer. Electrostatic repulsion from the negatively polarised electrode (at cathodic SWV potentials) may alter effective d_{eff} and hence k_{ET}. AAPVSGIG has no charged residues; IPRTLTAA must account for electrostatic correction to d_{eff}.

Layer 7: SAM formation and composition

Competitive thiol adsorption

Mixed SAM from probe-thiol (concentration c_p) and diluent-thiol (MCH or PEG-SH, concentration c_d). Equilibrium surface composition from competitive Langmuir:

$$\frac{\Gamma_p}{\Gamma_d} = \frac{c_p}{c_d} \cdot \frac{K_{a,p}}{K_{a,d}} \tag{25}$$

Short diluents adsorb preferentially ($K_{a,d} > K_{a,p}$): surface is diluent-enriched relative to solution. Solution composition must be tuned to achieve target $\Gamma_p = 1\text{–}10\,\text{pmol}\,\text{cm}^{-2}$.

Phase separation

In mixed SAMs, thiols can phase-separate into nanoscale domains (5–20 nm diameter). Probe-rich domains create local steric crowding even at low mean Γ_p. Domain formation is kinetically controlled; slow deposition or annealing promotes mixing. Must assume homogeneous SAM or account for effective local crowding.

Edinburgh calcium artefact

PEG spacer chelates Ca^{2+} with association constant $K_d \approx 1\text{–}10\,\text{mM}$ (literature for PEG-divalent cation complexes). At $[Ca^{2+}] = 1\,\text{mM}$ in wound fluid, occupied fraction $\theta_{\text{Ca}} = [Ca^{2+}]/(K_d + [Ca^{2+}])$. Complexed PEG contracts: effective change in MB-to-Au distance:

$$\Delta d_{\text{Ca}} = \Delta d_{\text{max}} \cdot \frac{[Ca^{2+}]}{K_d + [Ca^{2+}]} \tag{26}$$

This modifies k_{ET} (Layer 10). Since the scrambled control electrode uses identical PEG architecture, the Ca^{2+} artefact is the same on active and control probes and cancels in $F_k = 1 - \rho_{\text{active}}/\rho_{\text{control}}$, *provided* Γ_p is the same on both electrodes.

Fc reference SAM

C_6-Fc-SH with diluent. Keep $\Gamma_{\text{Fc}} < 3\,\text{pmol}\,\text{cm}^{-2}$ to avoid Fc-Fc through-space electron hopping (which distorts the SWV peak to an asymmetric shape). Diagnostic: symmetric SWV Fc peak confirms isolated molecules.

SAM hydration on first sample contact

The cartridge is stored dry (desiccated). When sample first contacts the SAM, the PEG spacer and peptide must hydrate before reaching their solution-state conformational equilibrium. PEG hydration occurs within ∼1–5 s; peptide equilibration within ∼1–10 s. If the $T = 0$ SWV scan is taken immediately after fill, probes may still be in a partially collapsed dry conformation, giving an anomalously high k^0_{ET} (shorter d_{eff} while dry) and elevated $R(0)$. Firmware specification: delay the $T = 0$ scan by ≥ 60 s after fill completion.

Wound redox agents and MB chemical reduction/oxidation

Co-eluted redox species from wound fluid alter the MB oxidation state at the electrode surface without applied potential — a signal artefact unrelated to enzyme cleavage.

Reducing agents (GSH ∼0.1–0.5 mM in chamber, ascorbate ∼0.05–0.2 mM):

$$\text{MB}^{2+} + 2\,\text{GSH} \rightarrow \text{MB} + \text{GSSG} + 2\text{H}^+, \quad k_{\text{red}} \approx 10\text{–}1000\,\text{M}^{-1}\text{s}^{-1} \tag{27}$$

At [GSH] = 0.5 mM and $k_{\text{red}} = 100\,\text{M}^{-1}\text{s}^{-1}$: pseudo-first-order half-life ≈ 14 s. MB converted to electroinactive leuco-MB within minutes of sample contact, reducing $R(0)$ independently of enzyme activity.

Oxidising agents (H_2O_2 10–100 μM, HOCl 1–100 μM): oxidise MB to its N-oxide (thionine oxide), also electroinactive. Net kinetics:

$$\frac{d[\text{MB}_{\text{active}}]}{dt} = -k_{\text{red}}[\text{GSH}][\text{MB}] - k_{\text{ox}}[\text{H}_2\text{O}_2][\text{MB}] \tag{28}$$

The ratiometric design does not automatically cancel this if the effect magnitude differs between active and control electrodes (probe sequence may influence local GSH access). Mitigation: include a redox-scavenging component in the elution buffer (e.g., catalase for H_2O_2), and model the residual error in F_k as a function of $[\text{GSH}]_{\text{chamber}}$.

Layer 8: Probe architecture and conformational dynamics

Probe structure

HNE channel: Ac-Cys-[$PEG_{4/6}$]-`AAPV`↓`SGIG`-[PEG_2]-Lys(MB)-NH_2

Contour lengths: $PEG_4 = 4 \times 0.36 = 1.44$ nm; $PEG_6 = 2.16$ nm; peptide (extended) $= 8 \times 0.35 = 2.8$ nm; $PEG_2 = 0.72$ nm. Total contour $L_c \approx 5.0$–6.0 nm.

End-to-end distance distribution

Worm-like chain model (Kratky-Porod) with persistence length l_p:

$$\langle r^2 \rangle = 2 l_p L_c \left[1 - \frac{l_p}{L_c} \left(1 - e^{-L_c/l_p} \right) \right] \tag{29}$$

For PEG: $l_p \approx 0.38$ nm; for unstructured peptide: $l_p \approx 0.4$–1.0 nm. The distribution of MB-to-Au distance $P(d)$ is broad. Because $k_{ET} \propto e^{-\beta d}$, the average ET rate:

$$\langle k^0_{ET} \rangle = \int_0^\infty k^{\max}_{ET} e^{-\beta d} P(d)\, dd \tag{30}$$

is dominated by the short-d tail of $P(d)$. Must compute numerically using the WLC distance distribution.

Steric accessibility

Maximum surface coverage allowing enzyme access: $\Gamma_{\text{steric,max}} \approx 1/(\pi r_E^2)$ where r_E is the enzyme radius. For MMP-9 ($r_E \approx 2.5$ nm): $\Gamma_{\text{steric,max}} \approx 5\,\text{pmol}\,\text{cm}^{-2}$. Keep $\Gamma_p < \Gamma_{\text{steric,max}}$.

Post-cleavage state

Cleavage at `AAPV`↓`SGIG`: remaining stub on surface is Ac-Cys-$PEG_{4/6}$-`AAPV` (no MB). Released fragment `SGIG`-PEG_2-Lys(MB)-NH_2 diffuses away. Only intact probes ($\Gamma_{\text{intact}} = \Gamma_0 - \Gamma_{\text{cleaved}}$) contribute to the MB SWV peak.

Layer 9: Enzyme kinetics on the surface

Surface Michaelis-Menten

$$\frac{d\Gamma}{dt} = -\frac{k_{\text{cat}}[E]_{\text{surface}}\,\Gamma}{K_{m,\text{surf}} + \Gamma} \tag{31}$$

where $[E]_{\text{surface}} = [E](z = 0, t)$ from the Layer 3 PDE, and $K_{m,\text{surf}} = K_{m,\text{sol}}/f_{\text{steric}}$ with steric correction factor $f_{\text{steric}} = \Gamma_{\text{steric,max}}/(\Gamma_{\text{steric,max}} + \Gamma_p)$ (effectively $K_{m,\text{surf}} > K_{m,\text{sol}}$ due to reduced accessibility).

First-order limit

When $\Gamma \ll K_{m,\text{surf}}$:

$$\frac{d\Gamma}{dt} \approx -\frac{k_{\text{cat}}}{K_{m,\text{surf}}}[E]_{\text{surface}}\Gamma \tag{32}$$

Solution (if $[E]_{\text{surface}} \approx$ const): $\Gamma(t) = \Gamma_0 \exp(-k_{\text{obs}}t)$ where $k_{\text{obs}} = (k_{\text{cat}}/K_{m,\text{surf}})[E]$. This is the regime where the assay measures k_{cat}/K_m.

Multi-protease competition

$$\frac{d\Gamma}{dt} = -\sum_j \frac{k_{\text{cat},j}[E_j]_{\text{surface}}\,\Gamma}{K_{m,\text{surf},j} + \Gamma} \tag{33}$$

Each interfering protease j (cathepsins, plasmin, etc.) contributes additional cleavage. Off-target contribution to F_k from protease j: $\delta F_k \approx (k_{\text{cat},j}/K_{m,\text{surf},j})[E_j]T$ at early times.

Cathepsin and plasmin cross-reactivity (MEROPS check)

The substrate predictor panel does not cover cathepsin G, cathepsin B, or plasmin. These must be assessed from MEROPS specificity data:

- **Cathepsin G** (serine protease): shares chymotrypsin fold with HNE; reports of Val at P1 being cleaved at ∼20–50% efficiency vs Phe. AAPV↓SGIG — P1 = Val is a moderate cathepsin G substrate. Risk: moderate.
- **Plasmin**: highly specific for Arg, Lys at P1. IPRT↓LTAA contains Arg at P3 in the sequence; if the scissile bond is I-P-R↓T, plasmin cleaves, releasing T-L-T-A-A-MB and leaving I-P-R on the electrode (no MB → false positive). Risk: high — requires MEROPS verification.
- **Cathepsin B**: endopeptidase specificity for Arg, Lys. AAPVSGIG has neither; IPRTLTAA has Arg at position 3. Moderate risk for the MMP-9 channel.

Wound fluid plasmin concentrations: ∼1–50 nM active form. Must compute $\delta F_k^{\text{plasmin}}$ for IPRTLTAA at [plasmin] = 10 nM and confirm $\delta F_k < 0.05$.

Layer 10: Marcus-theory electron transfer

Standard ET rate

$$k_{ET}^0(d) = A\exp(-\beta d)\exp\left(-\frac{\lambda}{4RT}\right) \tag{34}$$

where $\beta \approx 1.0\,\text{Å}^{-1}$ (flexible organic tether), $\lambda \approx 0.3$–$0.5\,\text{eV}$ (MB reorganisation energy), $A \approx 10^{12}$–$10^{13}\,\text{s}^{-1}$.

For $d = 3\,\text{nm}$: $k_{ET}^0 \approx 10^{13} \times e^{-30} \approx 10\,\text{s}^{-1}$. Measurable in SWV at $f \sim k_{ET}^0/(2\pi) \approx 2\,\text{Hz}$; increase f to 60 Hz to operate in quasi-reversible regime for a shorter effective tether.

Butler-Volmer potential dependence

$$k_f(\eta) = k_{ET}^0\exp\left(-\frac{\alpha F\eta}{RT}\right) \quad \text{(cathodic, reduces MB)} \tag{35}$$

$$k_b(\eta) = k_{ET}^0\exp\left(\frac{(1-\alpha)F\eta}{RT}\right) \quad \text{(anodic, oxidises MB)} \tag{36}$$

where $\eta = E_{\text{applied}} - E^\circ_{\text{MB}}$ and $\alpha \approx 0.5$.

Formal potential pH correction

$E^\circ_{\text{MB}} \approx -0.25\,\text{V}$ vs Ag/AgCl at pH 7.0. pH dependence:

$$E^\circ_{\text{MB}}(\text{pH}) = E^\circ_{\text{MB}}(7.0) - 0.059(\text{pH} - 7.0)\,\text{V} \tag{37}$$

At wound pH 6.5: $E^\circ_{\text{MB}} \approx -0.22\,\text{V}$; at pH 8.0: $E^\circ_{\text{MB}} \approx -0.34\,\text{V}$. A 120 mV shift across the operating range; must include pH correction in peak fitting.

Calcium artefact correction to d_{eff}

From Layer 7: $\Delta d_{\text{Ca}} = \Delta d_{\max} \times [\text{Ca}^{2+}]/(K_d + [\text{Ca}^{2+}])$. Modified ET rate:

$$k_{ET}^{0,\text{Ca}} = k_{ET}^{0}(d_0) \times \exp(\beta \Delta d_{\text{Ca}}) \tag{38}$$

Cancels in ratiometric F_k provided both electrodes have identical PEG architecture and Γ_p.

Temperature correction

Through Marcus theory: $k_{ET}^{0} \propto T^{1/2} \exp(-\lambda/(4k_B T))$. For a 15°C temperature range: k_{ET}^{0} can vary by a factor of 2–3. Include temperature as explicit parameter; ratiometric design partially corrects if temperature is uniform across the chip.

Layer 11: SWV forward model

SWV waveform

Square-wave voltammetry superimposes a symmetric square-wave (amplitude ΔE_{SW}, frequency f_{SWV}) on a staircase ramp (step ΔE_{step}). Net current: $\Delta I_n = I_{\text{forward,n}} - I_{\text{backward,n}}$ at each step.

Dimensionless kinetic parameter

$$\kappa_{\text{SW}} = \frac{k_{ET}^{0}}{f_{\text{SWV}}} \cdot \frac{RT}{nF} \tag{39}$$

Regimes: $\kappa_{\text{SW}} \gg 1$ (reversible, peak height $\propto \Gamma$, independent of k_{ET}^{0}); $\kappa_{\text{SW}} \approx 1$ (quasi-reversible, optimal); $\kappa_{\text{SW}} \ll 1$ (totally irreversible, peak flattens). Set f_{SWV} to achieve $\kappa_{\text{SW}} \approx 1$ for the MB probe.

Numerical forward model

At each potential step n with overpotential η_n, solve Butler-Volmer for surface coverage over one half-cycle ($\Delta t = 1/(2f_{\text{SWV}})$):

$$\frac{d\Gamma_{\text{ox}}}{dt} = k_b(\eta_n)\Gamma_{\text{red}} - k_f(\eta_n)\Gamma_{\text{ox}} \tag{40}$$

$$\Gamma_{\text{red}} = \Gamma_{\text{intact}} - \Gamma_{\text{ox}} \tag{41}$$

Net current per step: $\Delta I_n = nFA_{\text{true}}(d\Gamma_{\text{ox}}/dt|_{\text{forward}} - d\Gamma_{\text{ox}}/dt|_{\text{backward}})$.

Approximate peak current

For the quasi-reversible surface case:

$$I_{\text{peak,MB}} \approx \frac{n^2 F^2 A_{\text{true}} \Gamma_{\text{intact}} k_{ET}^{0} \Delta E_{\text{SW}}}{4RT} \tag{42}$$

MB and Fc peak separation

$\Delta E^\circ = E^\circ_{\text{Fc}} - E^\circ_{\text{MB}} = +0.45 - (-0.25) = 0.70\,\text{V}$. Peak FWHM for reversible surface wave: FWHM $= 5.56RT/(nF) \approx 65\,\text{mV}$ ($n = 2$, MB) and 131 mV ($n = 1$, Fc). Sum of half-widths $\ll 0.7\,\text{V}$: peaks are well-resolved. Verify numerically for quasi-reversible case.

Parameter optimisation

Sweep $(f_{\text{SWV}}, \Delta E_{\text{SW}}, \Delta E_{\text{step}})$ to maximise the ratio $I_{\text{peak,MB}}/I_{\text{cap}}$. Recommended starting point: $f = 60\,\text{Hz}$, $\Delta E_{\text{SW}} = 25\,\text{mV}$, $\Delta E_{\text{step}} = 2\,\text{mV}$. Scan window: -0.65 to $+0.65\,\text{V}$ vs Ag/AgCl.

Frequency diagnostic

If measured $I_{\text{peak}} \propto f^{0.5}$: diffusion-controlled (non-tethered MB in solution is dominating (wrong regime)). If $I_{\text{peak}} \propto f^1$ (at low f) or plateaus (at high f): surface-confined, correct. Must verify.

Dissolved oxygen interference

Air-equilibrated wound fluid and elution buffer carry $[\text{O}_2]_{\text{dissolved}} \approx 0.25\,\text{mM}$ at 25°C. On gold in neutral solution, O_2 reduction onset is kinetically shifted to $E \approx -0.3$ to $-0.5\,\text{V}$ vs Ag/AgCl — directly overlapping the MB peak at $E^\circ_{\text{MB}} \approx -0.25\,\text{V}$. The SWV scan to $-0.65\,\text{V}$ sits squarely in this window.

Estimated O_2 current at the MB peak potential:

$$i_{\text{O}_2} \approx \frac{4FD_{\text{O}_2}[\text{O}_2]}{\delta} \sim 2\,\text{nA} \tag{43}$$

for $A = 1\,\text{mm}^2$, $D_{O_2} = 2 \times 10^{-9}\,\text{m}^2\text{s}^{-1}$, $\delta = 100\,\mu\text{m}$. This is 5–20% of the MB faradaic peak (∼10–50 nA) and distorts the polynomial baseline assumed in Layer 12 peak extraction. The SWV net current partially suppresses the O_2 wave (diffusion-controlled processes contribute differently to forward and backward half-cycles), but does not eliminate it.

Mitigation options to model: (1) N_2 purge of elution buffer ($[O_2] \to\ < 0.01\,\text{mM}$, eliminates interference); (2) limit cathodic scan to $-0.45\,\text{V}$ (clips MB peak tail, may be acceptable); (3) electrochemical pre-depletion of O_2 before measurement scan. Option (1) is preferred and should be validated by comparing scans on purged vs unpurged buffer.

Multiple-scan kinetic protocol

If the protocol takes multiple SWV scans during incubation (e.g., $T = 0, 5, 10, 20, 30\,\text{min}$):

- **MB re-equilibration**: each scan drives MB through a complete redox cycle. Re-equilibration time $\tau_{\text{eq}} = 1/(k_f + k_b)$ at open-circuit potential is $\lesssim 1\,\text{s}$ for $k^0_{ET} \geq 10\,\text{s}^{-1}$. For inter-scan intervals $\geq 5\,\text{min}$, equilibration is complete; no scan-to-scan artefact.
- **O_2 depletion**: successive scans deplete local O_2 near the electrode, so later scans in a kinetic series show progressively less O_2 interference — must account for this trend in baseline fitting.
- **MB photodegradation**: MB is a photosensitiser; exposure to ambient light during incubation causes photobleaching at a rate $\propto$ light intensity. Device packaging must be opaque. If translucent, add first-order loss term $-k_{\text{photo}} I_{\text{light}} [\text{MB}_{\text{active}}]$ to the Layer 7.7 kinetic model.

Layer 12: SWV peak extraction and signal processing

Raw data processing

Typical scan: $(\Delta E_{\text{total}}/\Delta E_{\text{step}}) \approx 650$ points. Steps:

1. Savitzky-Golay smoothing (window = 11 pts, 3rd-order polynomial) to reduce high-frequency noise while preserving peak shape.
2. Cubic spline baseline through non-peak windows (e.g., -0.65 to $-0.45\,\text{V}$ and -0.05 to $+0.25\,\text{V}$ for MB channel).
3. Gaussian peak fit: $\Delta I(E) = A_{\text{peak}} \exp[-(E - E_p)^2/(2\sigma_E^2)]$ with parameters $(A_{\text{peak}}, E_p, \sigma_E)$ fitted by nonlinear least squares.
4. Peak area: $\mathcal{A} = A_{\text{peak}} \sigma_E \sqrt{2\pi}$ (integrated area is more robust to k_{ET}-frequency mismatch than peak height).

Quality diagnostics per scan

- Peak width $\sigma_E > 2\times$ expected ($> 65\,\text{mV}$ for MB): SAM heterogeneous or measurement error; flag.
- Peak position shift $|\delta E_p| > 5\,\text{mV}$ from scan to scan: reference drift or SAM degradation; flag.
- Fit residuals $> 3\sigma$: two-population MB (heterogeneous tether lengths) or non-Gaussian artifact; fit with two-Gaussian model.

Layer 13: Ratiometric signal and error propagation

Signal definitions

For electrode i at time T, using peak areas $\mathcal{A}_{\text{MB}}$ and $\mathcal{A}_{\text{Fc}}$:

$$R_i(T) = \frac{\mathcal{A}_{\text{MB},i}(T)}{\mathcal{A}_{\text{Fc},i}(T)}, \qquad \rho_i = \frac{R_i(T)}{R_i(0)}, \qquad F_k = 1 - \frac{\rho_{\text{active,k}}}{\rho_{\text{control,k}}} \tag{44}$$

$$\kappa_k = -\frac{\ln(1 - F_k)}{T} \tag{45}$$

Error propagation

$$\frac{\sigma_R}{R} = \sqrt{\left(\frac{\sigma_{\text{MB}}}{\mathcal{A}_{\text{MB}}}\right)^2 + \left(\frac{\sigma_{\text{Fc}}}{\mathcal{A}_{\text{Fc}}}\right)^2} \tag{46}$$

$$\frac{\sigma_\rho}{\rho} = \sqrt{2} \cdot \frac{\sigma_R}{R} \quad \text{(equal variance for } R(0) \text{ and } R(T)\text{)} \tag{47}$$

$$\sigma_F = \frac{\rho_a}{\rho_c} \sqrt{\left(\frac{\sigma_{\rho_a}}{\rho_a}\right)^2 + \left(\frac{\sigma_{\rho_c}}{\rho_c}\right)^2} \tag{48}$$

$$\sigma_\kappa = \frac{\sigma_F}{(1 - F_k)T} \tag{49}$$

Note: σ_κ diverges as $F_k \to 1$, setting the upper dynamic range ceiling.

Dynamic range

Useful range: $F_k \in [3\sigma_F,\ 0.90]$, corresponding to:

$$\kappa_k \in \left[\frac{3\sigma_F}{T},\ \frac{2.3}{T}\right] \tag{50}$$

Fouling cancellation and its limits

The double-ratio F_k cancels fouling that affects $\mathcal{A}_{\mathrm{MB}}$ equally on active and control probes. This holds if: (a) fouling is a bulk matrix effect (protein adsorption); (b) both electrodes have identical SAM diluent composition. If fouling is sequence-specific (albumin preferentially adsorbs to one peptide sequence), cancellation is incomplete. This assumption must be tested in the calibration experiment.

Layer 14: Electronics and potentiostat noise budget

TIA gain selection

Max expected current (all WEs, including capacitive): $I_{\max} \approx 200\,\mathrm{nA}$ per WE. For output voltage $< 1\,\mathrm{V}$: $R_f < 5\,\mathrm{M\Omega}$. Choose $R_f = 100\,\mathrm{k\Omega}$ (safety factor 10; noise scales as $1/\sqrt{R_f}$).

Noise sources

$$i_{\mathrm{Johnson}} = \sqrt{\frac{4k_BT \cdot \mathrm{BW}}{R_f}} \approx 4\,\mathrm{pA} \quad (R_f = 100\,\mathrm{k\Omega},\ \mathrm{BW} = 100\,\mathrm{Hz}) \tag{51}$$

$$i_{\mathrm{shot}} = \sqrt{2eI_{\mathrm{DC}}\,\mathrm{BW}} \approx 44\,\mathrm{fA} \quad (I_{\mathrm{DC}} = 60\,\mathrm{nA}) \tag{52}$$

$$i_{\mathrm{ADC}} = \frac{I_{\mathrm{range}}}{2^N\sqrt{12}} \approx 2\,\mathrm{fA} \quad (N = 24\,\mathrm{bit},\ I_{\mathrm{range}} = 100\,\mathrm{nA}) \tag{53}$$

Total: $i_{\mathrm{total}} \approx i_{\mathrm{Johnson}} \approx 4\,\mathrm{pA/Hz^{1/2}}$. Dominant noise source is Johnson noise in R_f.

Multiplexer settling

After switching WE: settling time $\tau = (R_{\mathrm{mux}} + R_s)C_{\mathrm{dl}}A \approx 33\,\Omega \times 2\,\mathrm{nF} = 66\,\mathrm{ns}$. Negligible vs SWV half-period of 8 ms.

Mains interference

50 Hz (EU mains) is close to SWV frequency $f = 60\,\mathrm{Hz}$. Beating frequency $|60 - 50| = 10\,\mathrm{Hz}$ creates slow modulation of the SWV baseline. Mitigation: shielded enclosure; or set $f_{\mathrm{SWV}} = 50\,\mathrm{Hz}$ synchronised to mains; or $f_{\mathrm{SWV}} \gg 100\,\mathrm{Hz}$ with appropriate κ_{SW} adjustment.

Required SNR

To resolve $F_{\mathrm{LOD}} = 5\%$: require $\sigma_F < 1.7\%$. With $\mathcal{A}_{\mathrm{MB}} \approx 1\,\mathrm{pA \cdot V}$ and $\sigma_{\mathcal{A}} \approx 0.4\,\mathrm{fA \cdot V}$ (from i_{Johnson} integrated over ~ 50 SWV points): $\sigma_{\mathcal{A}}/\mathcal{A} \approx 0.04\%$. Full propagation through $R \to \rho \to F$ gives $\sigma_F \approx 0.06\%$, well below the 1.7% target.

Layer 15: Calibration curve and limit of detection

Calibration model

In the first-order regime ($F_k \ll 1$):

$$F_k \approx \frac{k_{\mathrm{cat}}}{K_{m,\mathrm{surf}}}[E]_{\mathrm{chamber}}\,T \tag{54}$$

Linear in $[E]$; slope $= (k_{\mathrm{cat}}/K_{m,\mathrm{surf}}) \times T$ (from pES-01 to pES-08 wet lab + steric correction).

LOD and LLOQ

$$\mathrm{LOD} = \frac{3\sigma_{\mathrm{blank}}}{dF_k/d[E]}, \qquad \mathrm{LLOQ} = \frac{10\sigma_{\mathrm{blank}}}{dF_k/d[E]} \tag{55}$$

Expected: with $\sigma_F \approx 1\%$ and slope $\approx (10^4\,\mathrm{M^{-1}s^{-1}}) \times 1800\,\mathrm{s} = 1.8 \times 10^7\,\mathrm{M^{-1}}$:

$$\mathrm{LOD} \approx \frac{0.03}{1.8 \times 10^7} \approx 1.7\,\mathrm{nM} \tag{56}$$

Clinical MMP-9 range in wound fluid: 1–100 nM. Target LOD $< 1\,\mathrm{nM}$ after optimisation.

Two-channel deconvolution

Observed signal vector: $\mathbf{F} = [F_{\text{MMP9}}, F_{\text{HNE}}]^\top$. Sensitivity matrix:

$$\mathbf{F} = \mathbf{S} \cdot \mathbf{c}, \qquad \mathbf{S} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \tag{57}$$

where $S_{ij} = dF_{\text{ch},i}/d[E_j]$ (on-diagonal: specific; off-diagonal: cross-sensitivity). Estimated concentrations:

$$\hat{\mathbf{c}} = \mathbf{S}^{-1}\mathbf{F}, \qquad \sigma_{\hat{c}} = \mathbf{S}^{-1}\sigma_{\mathbf{F}} \tag{58}$$

Condition number $\kappa(\mathbf{S}) = \|\mathbf{S}\| \cdot \|\mathbf{S}^{-1}\|$ quantifies deconvolution noise amplification. Target: $\kappa(\mathbf{S}) < 30$ (same criterion as Panel A/B substrate selectivity). This directly links the substrate design dry-lab work to the device model: IPRTLTAA and AAPVSGIG were selected to minimise off-diagonal elements of $\mathbf{S}$, keeping $\kappa(\mathbf{S})$ low.

Layer 16: Full chain integration and falsifiable predictions

Forward simulation chain

Given inputs ($[E]_{\text{MMP-9,wound}}$, $[E]_{\text{HNE,wound}}$, T, pH, $[\text{Ca}^{2+}]$, T_{amb}):

1. Layer 0: ODE → active fractions at time of sampling
2. Layers 1–2: dilution chain → $[E]_{\text{chamber}}$
3. Layer 3: PDE (discretised, $N_z = 50$ grid points) → $[E](z = 0, t)$
4. Layer 7: competitive Langmuir → Γ_p, Γ_d
5. Layer 8: WLC model → $P(d)$ → $\langle k^0_{ET} \rangle$; steric factor
6. Layer 9: surface MM ODE (coupled to PDE) → $\Gamma_{\text{intact}}(t)$
7. Layer 10: Marcus, pH, Ca^{2+}, temperature corrections → $k^0_{ET,\text{eff}}$
8. Layer 11: SWV numerical model → $\mathcal{A}_{\text{MB}}(T)$, $\mathcal{A}_{\text{Fc}}(T)$
9. Layers 12–13: peak extraction, R, ρ, F_k, κ_k
10. Layer 14: add instrument noise
11. Layer 15: calibration lookup → $[\hat{E}]_{\text{MMP-9}}, [\hat{E}]_{\text{HNE}}$

Falsifiable outputs

- Predicted $F_k(T)$ vs measured $F_k(T)$ from calibration experiment in wound-fluid matrix.
- Predicted LOD vs measured LOD.
- Predicted $\kappa(\mathbf{S})$ vs measured cross-sensitivity (iGEM gold criterion: model predicts, wet lab confirms).
- Predicted Damköhler number regime vs observed F_k frequency dependence on $[E]$.

Global sensitivity analysis

Sobol variance decomposition (SALib) over all uncertain parameters to identify which layer contributes most variance to σ_{F_k}. Expected dominant parameters: Γ_p (SAM), $k_{\text{cat}}/K_{m,\text{surf}}$ (wet lab), d_{eff} (conformational model). Results will direct where to invest experimental effort to tighten the model.

Literature parameters required

Parameter	Symbol	Value	Source
MMP-9 diffusion coefficient	$D_{\text{MMP-9}}$	$\sim 5 \times 10^{-11}\,\text{m}^2\text{s}^{-1}$	Stokes-Einstein / experimental
HNE diffusion coefficient	D_{HNE}	$\sim 9 \times 10^{-11}\,\text{m}^2\text{s}^{-1}$	Stokes-Einstein / experimental
Tunnelling decay constant	β	$\sim 1.0\,\text{Å}^{-1}$	Smalley et al. 2003
MB reorganisation energy	λ	0.3–0.5 eV	Pheeney and Barton 2012
MB formal potential	E°_{MB}	−0.25 V (pH 7)	Kelley et al. 1997
Fc formal potential	E°_{Fc}	+0.45 V	Nowicka et al.
SAM double-layer capacitance	C_{dl}	1–5 $\mu\text{F cm}^{-2}$	Campuzano et al.
Optimal probe surface coverage	Γ_p	1–10 pmol cm^{-2}	Lubin and Plaxco 2010
PEG persistence length	l_p	~ 0.38 nm	Oesterhelt et al. 1999
Gold roughness factor	R_f	1.1–1.5	Electrochemical literature
Wound fluid $[\text{Ca}^{2+}]$		1–2 mM	Gethin et al.
Wound fluid pH range		6.5–8.2	Gethin et al.
k_{cat}/K_m MMP-9 on IPRTLTAA		from pES	pES-01 to pES-08 wet lab
k_{cat}/K_m HNE on AAPVSGIG		from pES	pES-01 to pES-08 wet lab

Implementation plan

1. **Layer 3 + 9**: implement coupled PDE-ODE in `models/03_fluidics/` and `models/05_surface_kinetics/` (Python, `scipy.integrate`). These are the critical kinetic layers.
2. **Layer 11**: implement SWV numerical forward model in `models/07_swv_forward/`. This is the single most important modelling component.
3. **Layer 8**: implement WLC distance distribution and compute $\langle k^0_{ET} \rangle$ in `models/06_electron_transfer/`.
4. **Layers 7, 14, 15**: SAM composition model, noise budget, calibration in their respective folders.
5. **Layer 16**: chain all layers into a single forward simulation in `models/16_forward_chain/forward_chain.py`.
6. **Sensitivity analysis**: SALib Sobol in `models/16_forward_chain/sensitivity.py`.
7. Populate `data/literature_parameters/` with structured CSV of all numeric parameters and DOI sources before implementing any layer.